Wheel-hub printed drive sprocket — Rev 011 — 11T on the UNCUT factory rim (no-cut twin)

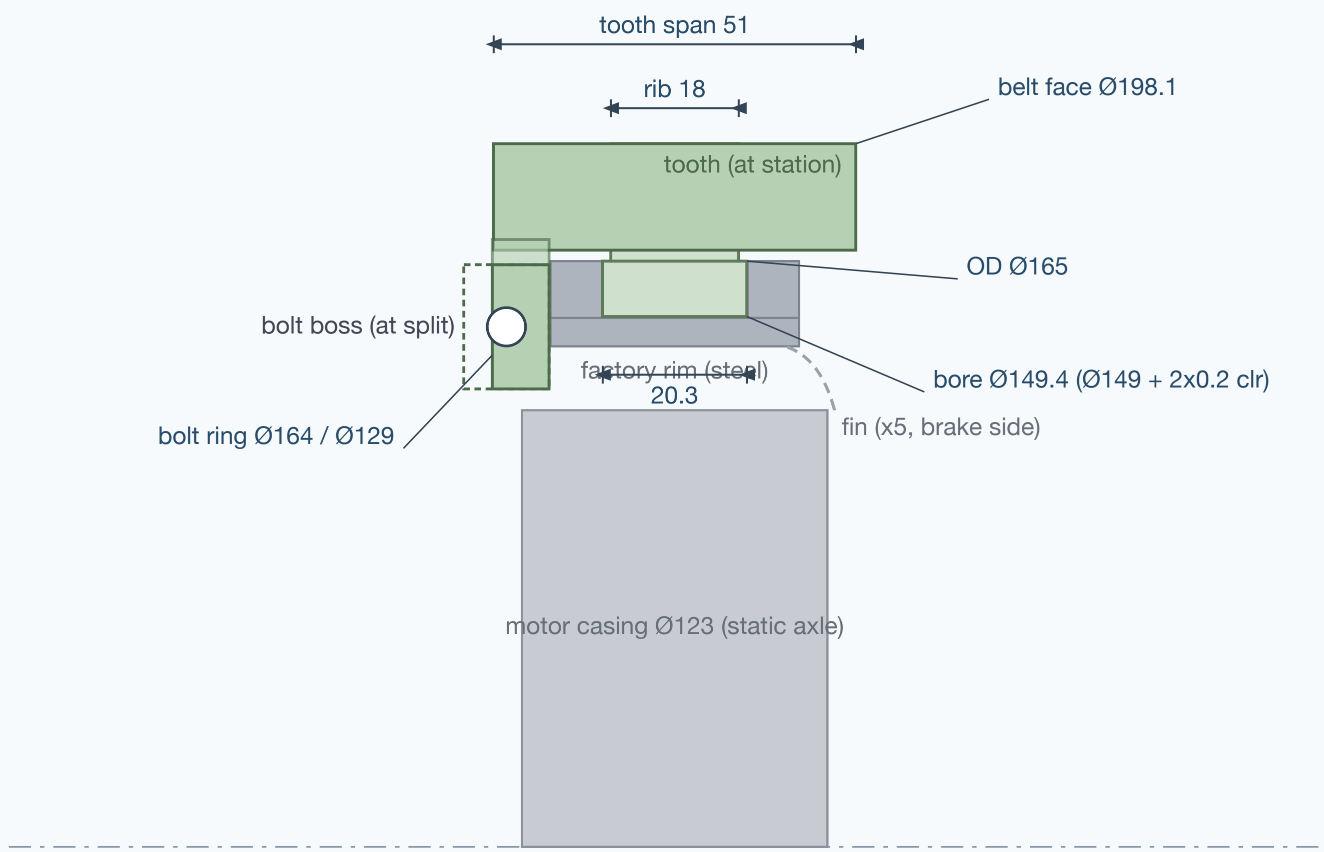
2026-07-28 · companion files: `sprocket_print.scad` (rev=9) · STLs: `stl/rev011_half_A.stl`, `stl/rev011_half_B.stl`, `stl/rev011_test_arc.stl` · geometry locked to `archive/rev011-uncut-11T`

What is new vs the assembly model: the .scad rev models draw the sprocket as rib + teeth only (visual). This blueprint adds the parts that make it a printable, installable component: the drum clamp well fill, the two-half split, and an under-floor bolt ring on the **non-fin side** so the halves bolt to each other through the open space between the hub's 5 fins. The bolt ring and bosses are a **new design addition** — verify fin clearance on the real wheel before printing (checklist below).

1 · Key dimensions

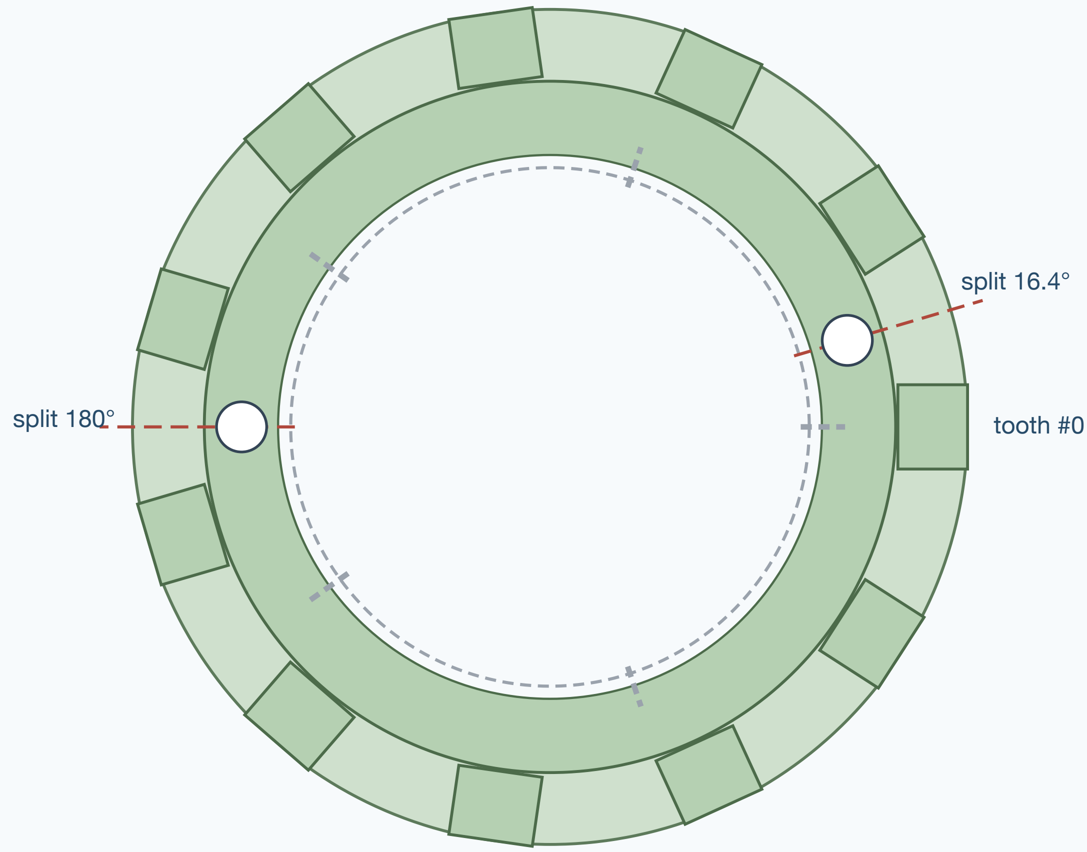
Teeth	11 T-teeth @ 32.73° — tooth marks every 56.6 mm on the rib top
Belt face (rib top)	Ø198.1 · cord circle Ø210.1 (belt: 60 mm pitch, 18 links, T = 12)
Drum seat	bore Ø149.4 = Ø149 floor + 2 x 0.2 fit clearance (tune with test arc) x 20.5 wide
Centre rib	18 wide x 16.54 tall — rides the 22 mm gap between the belt lug pairs
T-tooth	51 across x 20 thick, +5 root widening over the rib zone; outboard blades drop to Ø168.1 beside the rim
Well fill	bore Ø149.4 x 20.3 wide, OD Ø165 (flush with rim body)
Flange pockets	Ø165.8 x depth to |z| 17.75 — 0.4 radial / 0.1 axial clearance over the steel well walls (walls Ø149→Ø165, 7.25 wide each)
Bolt ring (NEW)	under-floor ring on the non-fin side: Ø129..Ø164 x 8 thick, registers 0.2 off the rim face; tied into every tooth blade
Split	bent split (11T is odd): joint lines at 16.4° and 180.0° — halves are 163.6° / 196.4° arcs, both joints still fall in tooth gaps · 0.25° trimmed per face (~0.3 mm) so the clamp can close
Hardware	2 x M5 x 35 cl.8.8 + nyloc + washers — one per joint line, tangential through the boss pairs (Ø5.4 holes)
Clearances (guarded)	ring OD to belt-lug tip sweep (Ø168.1): ≥2 mm · ring ID to motor casing (Ø123): 3 mm

2 · Section



half-section through a tooth station · ring side (no fins) on the left · steel = gray ghost, print = green

3 · Face view, split & clocking



face view from the ring side · 11 teeth @ 32.73° (split lines fall in tooth gaps)
fins (dashed) on a 72° grid — verify both bolt bosses land between fins before printing

4 · Printing

- **Material:** ABS (or ASA). **Orientation:** axis vertical, **bolt-ring face DOWN** — teeth print as continuous layer planes (per the project README).
- **Strength:** ≥4 perimeters, ≥40% infill (gyroid/cubic), 0.2 layers; no cooling fan for layer adhesion (ABS).
- **Supports:** needed under the fill edge and rib overhangs between tooth stations (the connectors support the part at each tooth).
- **Fit:** `bore_clr = 0.2` per side is the starting value — print the TEST ARC first and adjust for your printer's ABS shrinkage before committing to the halves.

5 · Assembly

1. Degrease the drum seat (the tire well).
2. Dry-fit both halves; the split faces should show a small even gap (~0.3 mm/side) — that gap is the clamp stroke.
3. Clock tooth #0 to your reference mark on the hub; confirm both bolt bosses sit between fins.

4. Bolts with washers both sides, nyloc nuts; tighten alternately until the gap closes evenly. Do not crush — snug + a quarter turn (ABS creeps; re-check after the first ride).
5. Spin check: rib runout ≤ 0.5 mm; tooth marks 56.6 mm apart on the rib top.

6 · Pre-print checklist (mandatory)

- Verify well floor Ø149 x 20.5 and flange height 8 (Ø165 body) on the actual rim
- Print the TEST ARC first — seat it in the well; tune bore_clr until snug
- Verify the 5 fins sit at 72.0° nominal and the two bolt bosses (16.4° / 180° from tooth #0) land between them
- Tooth spacing check on the print: marks every 56.6 mm on the rib top (32.73°)